

Discussion 3-Counting

1- You are given the set of letters $\{A, B, C, D, E\}$. How many five-letter strings can be made if we require that each letter appears exactly once and the letters A and B are next to each other, as either “AB” or “BA”?

2- Ms. Jones has 10 books that she is going to put on her bookshelf. Of these, 4 are mathematics books, 3 are chemistry books, 2 are history books, and 1 is a language book. Ms. Jones wants to arrange her books so that all the books dealing with the same subject are together on the shelf. How many different arrangements are possible?

3- From a group of 5 women and 7 men, we want to choose a committee of 5 people.

- (a) Probability of choosing 3 women and 2 men?
- (b) Probability of choosing 3 women and 2 men if 2 of the men are feuding and refuse to serve on the committee together?

- 4. a. In how many ways can 3 boys and 3 girls sit in a row?
- b. In how many ways can 3 boys and 3 girls sit in a row if the boys and the girls are each to sit together?
- c. In how many ways if only the boys must sit together?
- d. In how many ways if no two people of the same sex are allowed to sit together?

5. If n people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year? For this question assume a year has 365 days.

6. Suppose $P(B) = 0.45$, $P(A \cup B) = 0.67$, $P(A \cup B') = 0.89$. Calculate the followings

- (a) $P(A)$
- (b) $P(A|B)$
- (c) $P(A|B')$